Foundations NRSG 231 · Midterm Study Guide (v4.5)

Midterm: Wed Oct 21, 2026, 8:30 AM, Room B137. Examplify, 90 min, 50 points. Question types: multiple choice, short answer, select all that apply, hot spot (click on a photo) and drag and drop. Arrive 15 min early and bring your phone and name badge.

Built only from her materials: Ch. 25 Asepsis slides, the Ch. 34 Activity lecture, Ch. 26 Vital Signs slides, Ch. 8 Communication slides, the Communication Scenario role-play deck, Weekly Skills FALL 26, her **Vital Sign Cheat Sheet**, the three **Vital Signs Extra Practice** handouts, the **BP Practice** PPT, and the Midterm FAQ "Key Content Areas" list. There is **no textbook content**. Organized by the FAQ topics, in FAQ order.

NOT IN SLIDES marks an FAQ item her slides don't answer. Get it from class notes or the lab checklist; this guide doesn't fill it in. The FAQ itself says **"MEMORIZE the vital sign cheat sheet."**

✅ Exam FAQ checklist (from her Exam FAQ)

- **When:** Wed 10/21, 8:30 AM. **Arrive 15 min early. Where:** Room B137.
- **Worth:** 50 pts. **Time:** 90 min.
- **Question types:** multiple choice, short answer, select all that apply (SATA), hot spot, drag-and-drop.
- **MEMORIZE the vital sign cheat sheet** (next box).
- **Chairs face the back** of the room.
- **Covers:** Module 1 Ch. 25 Asepsis · Module 2 Ch. 34 Safe Patient Handling · Module 3 Vital Signs pt 1 & 2 · Module 4 Communication pt 1.
- **Also on the exam: Medical Terminology (Ehrlich)**: Ch. 1 word parts, Ch. 2 body directions, Ch. 5 blood terms, Ch. 6 immune terms (Med Term Quizzes #1–#2). See the purple “Medical Terminology (Ehrlich)” section.

★ MUST MEMORIZE: NRSG 231 Vital Signs Ranges Cheat Sheet (her handout, verbatim)

Blood Pressure (BP)

- Normal: Less than 120/80
- Low BP (Hypotension): Below 90 (systolic) or below 60 (diastolic)
- Elevated (Prehypertension): 120–129 (systolic) and less than 80 (diastolic)
- High BP Stage 1: 130–139 / 80–89
- High BP Stage 2: 140+ / 90+
- Hypertensive Crisis: Over 180 / Over 120 Emergency

Temperature

- Normal: 96.4°F – 100.4°F (Think: 37°C = heaven)
- Fever (Pyrexia): Over 100.4°F (38°C = not so great)
- Hypothermia: Below 95°F (35°C = barely alive)
- Hyperthermia: Over 104°F (40°C)
- Hyperpyrexia (Super Fever): Over 106°F (41°C) Medical Emergency

Heart Rate (HR) – Beats per Minute (BPM)

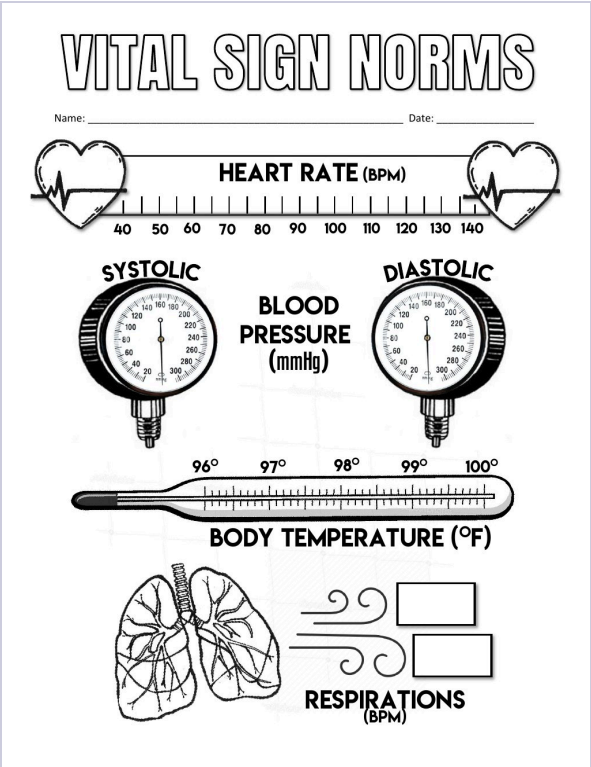
- Normal: 60–100 BPM
- Bradycardia (Too Slow): Less than 60
- Tachycardia (Too Fast): More than 100

Respiratory Rate (Breaths per Minute)

- Normal: 12–20 breaths/min
- Bradypnea (Too Slow): Less than 12
- Tachypnea (Too Fast): More than 20

Oxygen Saturation (SpO_2)

- Normal: 95%–100%
- Borderline Low: 92%–94%
- Hypoxia (Too Low): Below 92%



Vital Signs Extra Practice 2 handout: "Vital Sign Norms" (HR, BP, temp, RR)

Cheat sheet vs her slides (both are hers; the FAQ says to memorize the cheat sheet):
• Elevated BP: cheat sheet says diastolic **less than 80**; Slide 20 says diastolic "of 80." The cheat sheet settles the overlap.
• Stage 2: cheat sheet **140+/90+** (includes 140/90); Slide 20 says ">140, >90."
• Crisis: cheat sheet **over 180 / over 120**; Slide 20 gives only ">180."
• Normal temp: cheat sheet **96.4–100.4 °F**; Slide 5 says cells work best at 36–37.5 °C (98.6 °F).
• Hyperthermia: cheat sheet **over 104 °F (40 °C)**; Slide 7 calls >40 °C (104 °F) "dangerous." Hyperpyrexia >106 °F (41 °C) matches.

1 · Ch. 25 Asepsis & Infection Control

Chain of infection and how to break it [Slides 3, 4, 14, 16]

Infection = a disease state from pathogens in or on the body. All **six links** must be present [Slide 3]:

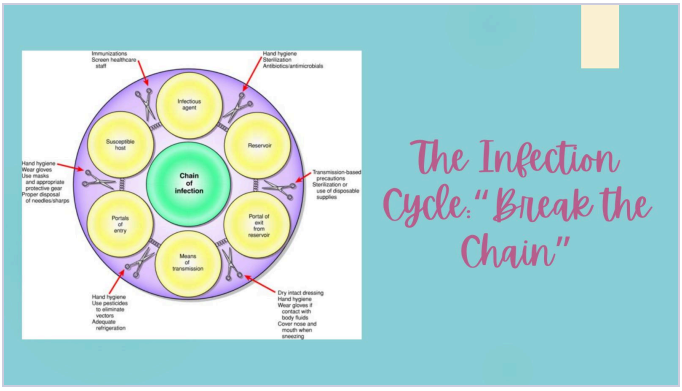
Link	Her examples	Break the chain [Slide 4 figure]
1. Infectious agent	Bacteria (E. coli, staph, strep); viruses (HIV, cold, flu, COVID-19, hepatitis A/B/C); fungi (athlete's foot, ringworm, yeast); parasites (malaria via mosquito) [Slide 5]	Hand hygiene, sterilization, antibiotics/antimicrobials
2. Reservoir	People, animals, soil, food, water, milk, objects [Slide 6]	Transmission-based precautions; sterilization or disposable supplies
3. Portal of exit	Respiratory, GI, GU tracts, skin breakdown, blood & drainage [Slide 11]	Dry intact dressing, hand hygiene, gloves if contact with body fluids, cover nose and mouth when sneezing
4. Means (mode) of transmission	**Contact**: direct (touching, kissing, sex) or indirect via a **vector** (living creature) or **fomite** (inanimate object) [Slide 8]. **Droplet**: organisms >5 µm, heavy, land on surfaces. **Airborne**: organisms <5 µm, stay in the air [Slide 9]	Hand hygiene, pesticides to eliminate vectors, adequate refrigeration
5. Portal of entry	Respiratory, GI, GU tracts, skin breakdown, blood & tissues [Slide 7]	Hand hygiene, gloves, masks and protective gear, proper sharps disposal
6. Susceptible host	Multiple wounds/poor skin integrity (= multiple portals of entry), pH of GI/GU/skin, WBC count, age/sex/heredity, immunity (natural vs acquired vs passive), fatigue, nutrition, pre-existing illness, certain meds, stress, invasive or indwelling devices [Slide 12]	Immunizations; screen health care staff

Slide 14 (APIC poster), 5 steps: 1 Stop at the source · 2 Block the portal of exit · 3 Interrupt mode of transmission · 4 Protect portal of entry · 5 Increase host's defenses.

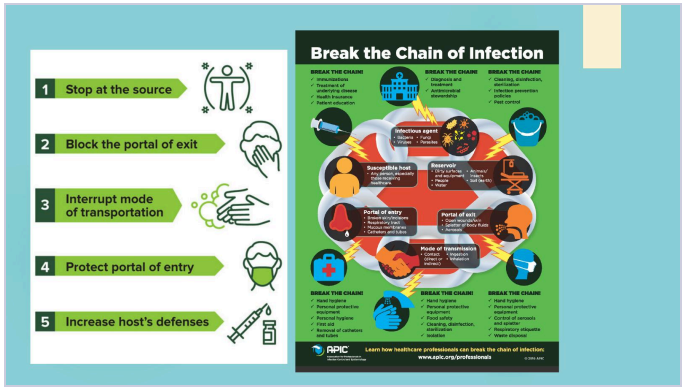
Her practice question [Slide 16]: breaking the chain at the *mode of transmission* = **wearing gloves during wound care**. Antibiotics act on the agent, and nutrition strengthens the host.

Case study, Jimin [Slide 13]: leg laceration, IV in his arm, unvaccinated for COVID, dehydration, stress and fatigue all make him a susceptible host.

What makes an organism thrive [Slide 15]: quantity, virulence, the patient's immune system, how long the contact lasted, and what type of contact it was.



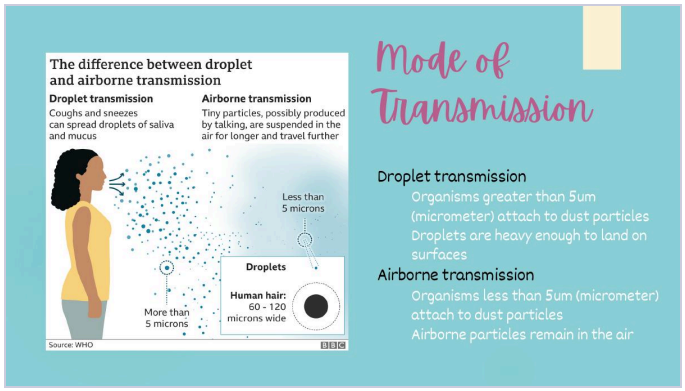
Slide 4: break the chain



Slide 14: APIC poster



Slide 8: modes of transmission (how germs spread)

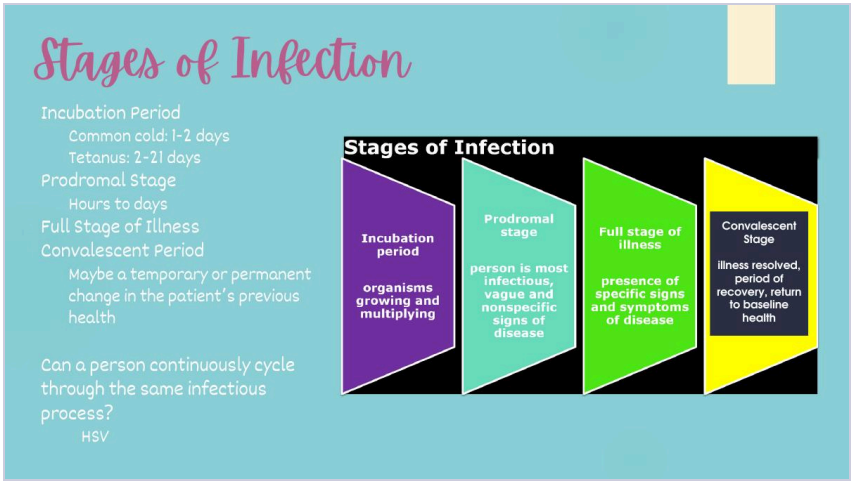


Slide 9: droplet vs airborne transmission

Stages of infection [Slides 17–18]

1. **Incubation**: e.g. common cold 1–2 days, tetanus 2–21 days
2. **Prodromal**: hours to days; vague symptoms
3. **Full stage of illness**
4. **Convalescent**: illness resolves, recovery, return to baseline. There may be a temporary or permanent change from previous health.

Can someone cycle through the same infection again? Yes, e.g. **HSV** [Slide 17]. **Her knowledge check [Slide 18]:** a student who is "too tired to keep my head up" with a low-grade fever is in the **prodromal** stage.



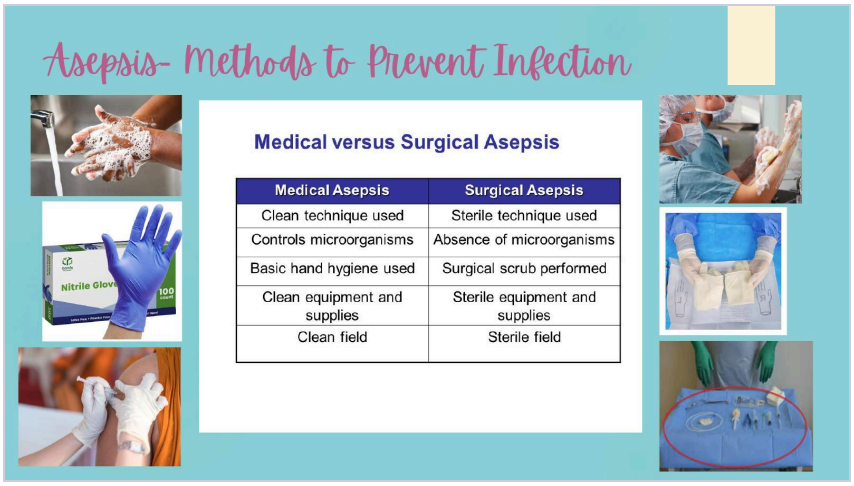
Slide 17: stages of infection

Medical asepsis vs surgical asepsis, and asepsis practices [Slides 19–21]

Medical asepsis: "clean technique"	Surgical asepsis: "sterile technique"
Reduces the number of pathogens; controls microorganisms	**Eliminates** pathogens; absence of microorganisms
Basic hand hygiene	Surgical scrub
Clean equipment and supplies; clean field	Sterile equipment and supplies; sterile field
Practices: hand hygiene, gloves, never put soiled linen on the floor, avoid raising dust, clean-to-dirty, put dirty linen directly in the hamper, pour contaminated liquids into the drain or "hopper" to avoid splashing	Creates and keeps a pathogen-free environment. Required for Foley catheters and central line placement/dressing changes. Sterile gloves, mask, hair net, gown, etc.

WHO Five Moments for Hand Hygiene [Slide 21]: 1 Before touching a patient · 2 Before a clean or aseptic procedure · 3 After body fluid exposure risk · 4 After touching a patient · 5 After touching patient surroundings.

NOT IN SLIDES Slide 24 only *asks* how long to wash with soap and water, how to use sanitizer, which is more effective and when sanitizer isn't appropriate. The answers aren't on the slides.



Slide 20: medical vs surgical asepsis



Slide 21: WHO Five Moments for Hand Hygiene



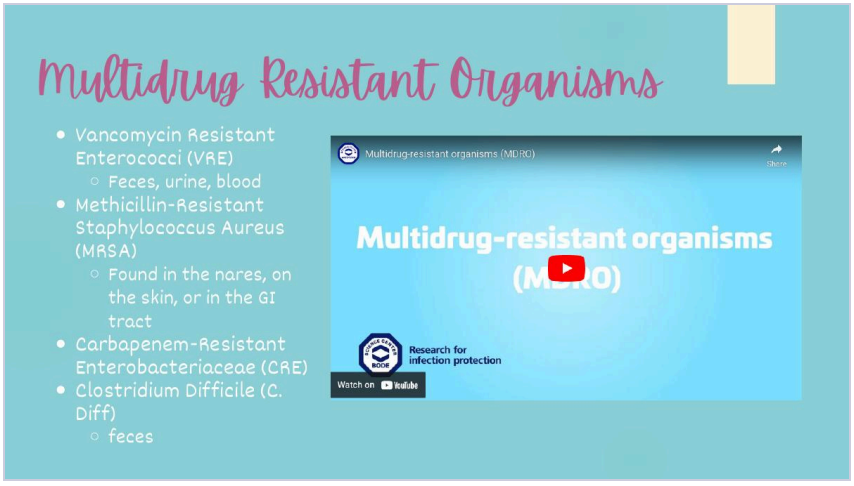
Slide 33: hand hygiene

MDROs: what they are and how to prevent them [Slides 23, 29]

- **VRE** (vancomycin-resistant Enterococci): feces, urine, blood
- **MRSA** (methicillin-resistant Staph aureus): nares, skin, GI tract
- **CRE** (carbapenem-resistant Enterobacteriaceae)

- **C. diff** (Clostridium difficile): feces

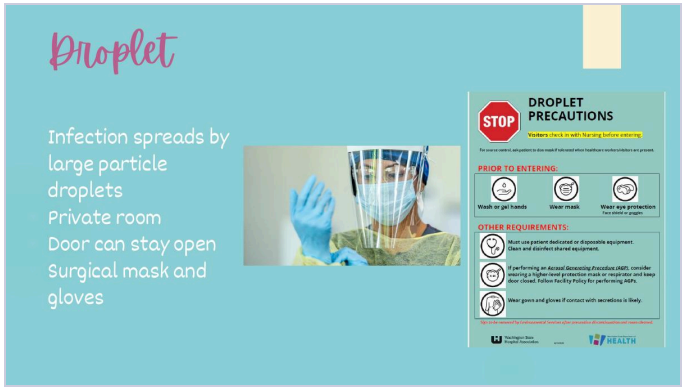
Prevention on her slides: a patient infected or colonized with an MDRO goes on **contact precautions**, with a private room, gown and gloves, glove changes after touching infective material, and equipment cleaned and not shared [Slide 29].



Slide 23: multidrug-resistant organisms (MDROs)

Isolation precautions [Slides 26–31]

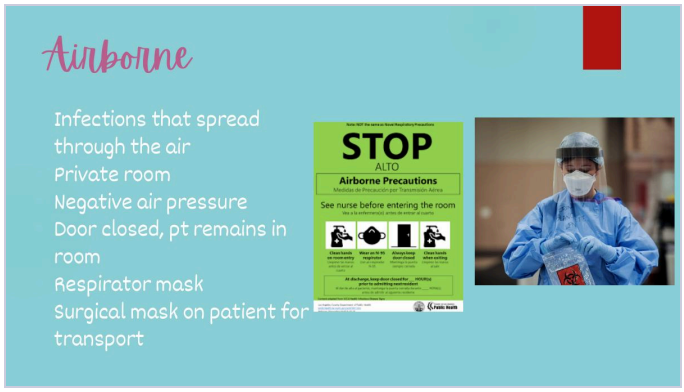
Type	Her slide details
Standard [26]	All hospitalized patients regardless of diagnosis. Applies to blood, all body fluids, secretions and excretions **except sweat**, non-intact skin and mucous membranes. Newer additions: respiratory hygiene/cough etiquette, safe injection practices, and a mask for high-risk prolonged spinal canal procedures.
Transmission-based [27]	Used **in addition to** standard precautions for suspected airborne, droplet or contact pathogens. Don PPE when entering, doff when leaving. A disease can have more than one route.
Droplet [28]	Large-particle droplets · private room · **door can stay open** · surgical mask and gloves
Contact [29]	MDRO · private room · **gown and gloves** · change gloves after infective material · clean equipment, don't share
Airborne [30]	Spread through the air · private room · **negative air pressure** · **door closed**, patient stays in the room · **respirator mask** · surgical mask on the patient for transport
Reverse/neutropenic [31]	Severely immunocompromised/immunosuppressed; any cancer treatment or transplant; **WBC below 500 cells/mm³**



Slide 28: droplet precautions



Slide 29: contact precautions



Slide 30: airborne precautions



Slide 31: reverse/neutropenic precautions

PPE [Slides 25, 27; Weekly Skills Week 2]

PPE = **gloves, gown, masks, protective eyewear** [Slide 25]. Don PPE entering a contact or droplet room and doff when leaving [Slide 27]. Week 2 lab and ATI checklists: Performing Hand Hygiene, **Applying PPE, Removing PPE**; TracPrac: standard precautions with hand hygiene, **Don and Doff PPE** [Skills p. 1].

NOT IN SLIDES The donning and doffing **order** isn't on her slides or the skills sheet. Use the ATI "Applying PPE / Removing PPE" checklist from Week 2 lab.



Slide 25: PPE = gloves, gown, masks, protective eyewear

DIAGRAM · PPE order (standard CDC sequence, not on her slides)

DON (put on) when entering

1	2	3	4	5
Hand hygiene	Gown	Mask / respirator	Goggles / face shield	Gloves (over gown cuffs)

DOFF (take off) at the door when leaving

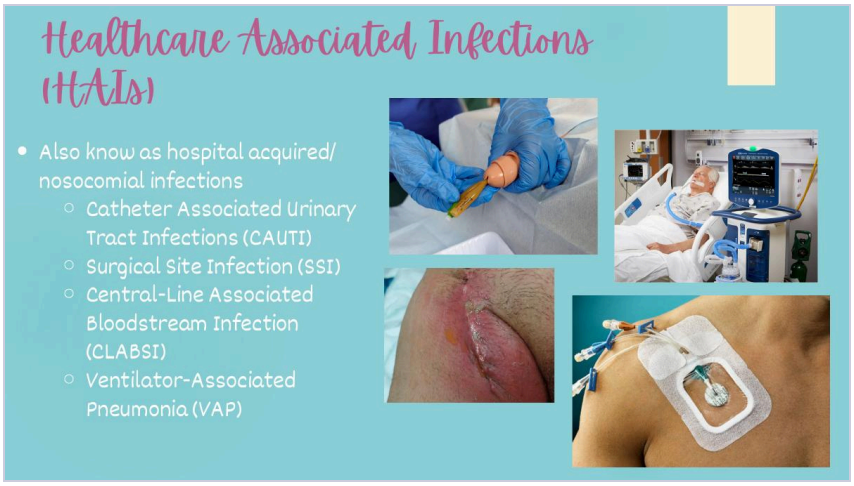
1	2	3	4	5
Gloves (most dirty)	Goggles / face shield	Gown	Mask / respirator	Hand hygiene

Confirm this order against your ATI "Applying PPE / Removing PPE" checklist from Week 2 lab.

Diagram (drawn, not from her materials): standard CDC donning/doffing order. Her slides and skills sheet don't give the order; check it against the ATI checklist.

HAIs: what they are and how to prevent them [Slide 22]

Health care–associated infections, also called hospital-acquired or nosocomial infections: **CAUTI** (catheter-associated UTI), **SSI** (surgical site infection), **CLABSI** (central line–associated bloodstream infection), **VAP** (ventilator-associated pneumonia).



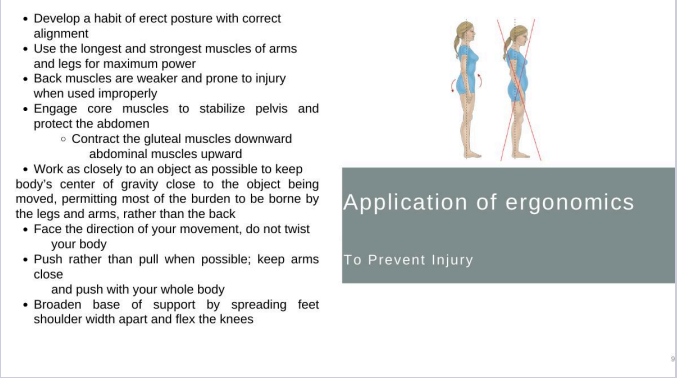
Slide 22: healthcare-associated infections (HAIs)

Tie-ins from her slides: indwelling devices make a patient more susceptible [Slide 12]. Foley catheters and central lines require **surgical asepsis** [Slide 19]. Slide 24 asks what you can conclude when a patient with a device shows infection signs. **NOT IN SLIDES** She doesn't give an HAI-specific prevention list beyond those points. Her answer to "best way to prevent infection?" (Slide 32) is shown only as a question, but the slides keep coming back to hand hygiene.

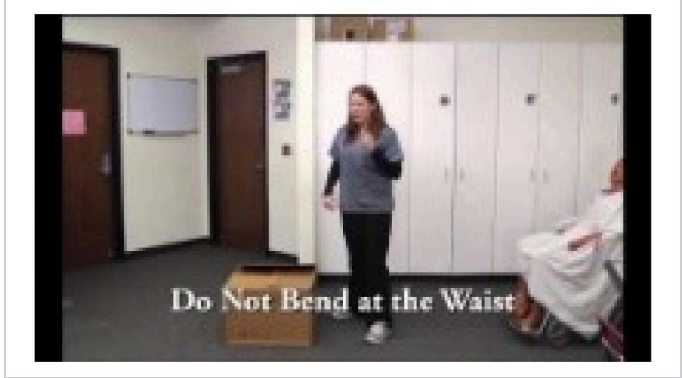
2 · Ch. 34 Safe Patient Handling (Activity lecture + Week 3 lab)

Positioning & body mechanics [Slides 3–7, 12]

- **Alignment/posture**: best musculoskeletal balance with no undue strain on joints, muscles, tendons or ligaments. **Balance**: center of gravity close to a wide base of support. Spreading the feet and flexing the hips and knees increases balance and reduces strain [3].
- **Coordinated movement** uses the major, stronger muscle groups. **Postural reflexes** (inner ear, vision, autonomic contractions) maintain position and equilibrium [4].
- Factors affecting movement: developmental level; physical health (hip dysplasia, torticollis, scoliosis; dwarfism, osteoporosis, muscular dystrophy; arthritis, trauma; stroke, spinal cord injury); mental health (depression); lifestyle (diet, sedentary, smoking); culture. ADLs = eating, bathing, dressing, toileting [5].
- **Ergonomics**: designing equipment and tasks to fit the worker so injuries are prevented [6].
- **Preventing injury [7]**: erect posture; use the **longest, strongest muscles of the arms and legs** (the back is weaker); engage the core (glutes down, abs up); **work close to the object**; **face the direction of movement, don't twist**; **push rather than pull**, arms close; feet shoulder-width apart with knees flexed. Video: **do not bend at the waist** [8].

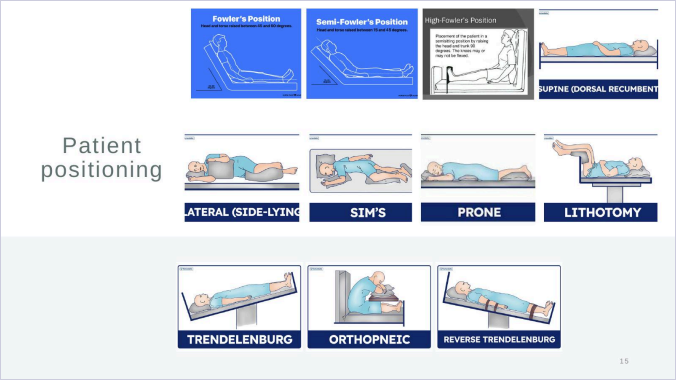


Slide 7: preventing injury (posture, body mechanics)



Slide 8 video: "Do not bend at the waist"

Positions [Slide 12]: Fowler's (head of bed 45–60°), semi-Fowler's (15–45°), high-Fowler's (about 90°, knees may or may not be flexed), supine (dorsal recumbent), lateral (side-lying), Sim's, prone, lithotomy, Trendelenburg, orthopneic, reverse Trendelenburg.



Slide 12: patient positioning

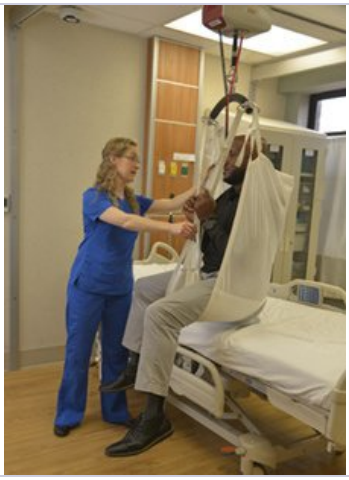


Slide 11: mobility devices

Ensuring safe patient handling [Slides 9–11]

1. Assess the patient: ability to understand and cooperate, diagnosis and capabilities, braces/devices; encourage them to help as much as they can
2. Enough staff present
3. Clear clutter and check access and devices
4. Use a screening/assessment tool
5. Choose the equipment
6. Plan and communicate with the team and the patient
7. Give a prescribed analgesic before the move if the patient is in pain
8. Elevate the bed as needed
9. **Lock the wheels**
10. Keep the patient safe: avoid friction, support good alignment (don't grab), move in smooth rhythmic motions

Causes of handling injuries [10]: uncoordinated lifts, high exertion in awkward positions, manual lifting without devices, repetitive lifting and repositioning, lifting while fatigued, standing for long periods, bed-to-gurney/wheelchair transfers, repositioning in bed, and patients who are cognitively impaired, dependent, obese or unable/unwilling to help. **Devices [11]:** friction-reducing sheets, standing assist, gait belts, lateral assist device, powered full-body lifts.



Activity Slide 6: powered lift for safe patient handling



Activity Slide 6: lateral transfer with a team

FAQ lab skills [Slide 13; Weekly Skills Week 3]

Week 3 lab: ambulation/assisted fall, wheelchairs, positioning/Hoyer lift, lateral transfers. ATI checklists: Moving and Turning Clients in Bed, Safe and Effective Transfer Techniques, Transferring Clients. TracPrac: ambulate a client; transfer to chair, commode or wheelchair; reposition in bed with safe body mechanics [Skills p. 1–2].



Activity Slide 3: pivot transfer with a gait belt and non-slip shoes

NOT IN SLIDES Step-by-step procedures for **ambulation with safe (assisted) fall, wheelchair transfer, lateral transfer to gurney and repositioning in bed** aren't on the slides. Slide 13 only lists them. Use the ATI checklists and lab notes. *Steps: see “Not from the professor — double-check” at the end.*

3 · Vital Signs Pt. 1 & 2

The vital signs are temperature, pulse, respirations, BP, pulse oximetry and pain level [Slide 3]. **When to assess [4]:** on admission; per policy; any change in condition; loss of consciousness; before and after surgery or invasive procedures; before and after risky activity (e.g. first ambulation after surgery); before meds that affect heart or lungs; whenever the patient reports a change, pain or distress. A baseline set lets you spot changes, and a client's own normal can differ from standard values [26].

Temperature pathophysiology [Slides 5–10]

- **Hypothalamus** controls thermoregulation by conserving heat or increasing heat loss. Temperature = heat produced minus heat lost [5].
- Affected by time of day, biological sex, age, activity, health and environment. **Core** sites: intracranial, intrathoracic, intra-abdominal. Cells work best at **36–37.5 °C (98.6 °F)** [5].
- **Circadian**: lower in the morning, **peaks 1500–1800**. Extreme cold leads to hypothermia and extreme heat to hyperthermia [6]. Slide 6 asks whether older adults run higher or lower and about women; **NOT IN SLIDES** the answers aren't given.
- **Fever (pyrexia)**: febrile vs afebrile. **>40 °C (>104 °F) dangerous**; **≥41 °C (106 °F) = hyperpyrexia, a medical emergency**. Signs: loss of appetite, headache, hot dry skin, thirst, muscle aches, fatigue [7]. Slide 8 asks which other vital sign rises with fever; **NOT IN HER MATERIALS** the answer isn't given.
- **Treating fever [9]**: fever is an immune defense, so treatment aims for comfort and preventing complications. Drugs: antibiotics for bacterial infection, antipyretics. Non-drug: cool bath, cooling blankets, removing extra clothing and blankets, oral fluids.
- **Hypothermia [10]**: poor coordination, slurred speech, amnesia, hallucinations, stupor; RR, HR and BP **decrease**. Treat by raising body temperature: warm blankets, warm fluids, Bair Hugger.



Slide 7: fever (pyrexia)



Slide 10: treating hypothermia

All methods to measure temperature [Slides 11–12; Skills p. 2]

Route	Key points	Time
Oral	Posterior sublingual pocket (near the carotid artery). No hot/cold drinks or smoking for 20 min before. Patient must be awake and alert. Not for small children (they may bite down).	3 min
Axillary	Bulb in the center of the axilla, lower arm across the chest. Non-invasive and good for children but less accurate. **0.5 °C lower than oral.**	5–10 min

Rectal	Side-lying with the upper leg flexed; lubricated, inserted 1–1½ in (adult). Use when oral is unsafe or inaccurate (unconscious, disoriented, irrational). **0.5 °C higher than oral.**	2–3 min
Tympanic (ear)	Close to the hypothalamus, so sensitive to core changes. **Adult: pull the pinna up and back.** Fast and easy. Cerumen impaction or otitis media distorts the reading.	2–3 sec

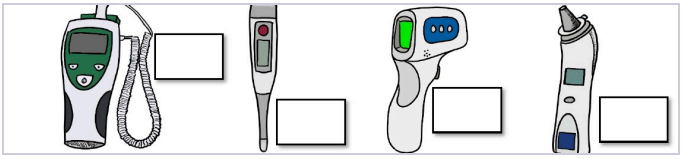
Thermometer types: infrared, tympanic, electronic oral/axillary, electronic rectal attachment [11]. The Week 4 skills sheet also lists a **temporal** video and says rectal is "discussion" only. **Extra Practice 3 handout:** temporal = taken on the **forehead**; axillary = "arm pit" with the arm in **adduction**; tympanic = **external auditory canal**; oral = **sublingual pocket using the sublingual artery**; rectal = in the rectum. The handout also asks which site is most accurate for core temp, which is affected by recent food and fluids, which shows rapid changes (near the hypothalamus), which is "non-touch," and which is least reliable. It gives no answer key, so use the slide facts above (tympanic is near the hypothalamus; no hot/cold drinks for 20 min before oral). **NOT IN HER MATERIALS** Temporal step-by-step technique.



Slide 11: thermometers



Slide 12: sites for assessing temperature



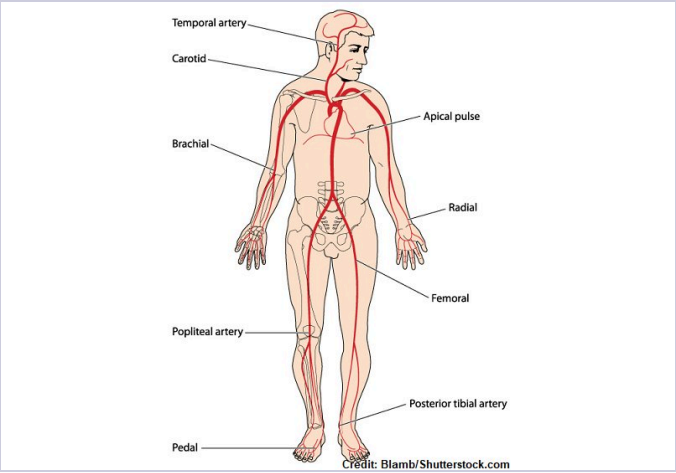
Extra Practice 3 handout: thermometer types to match to their route

First-quarter scope for abnormal temperature

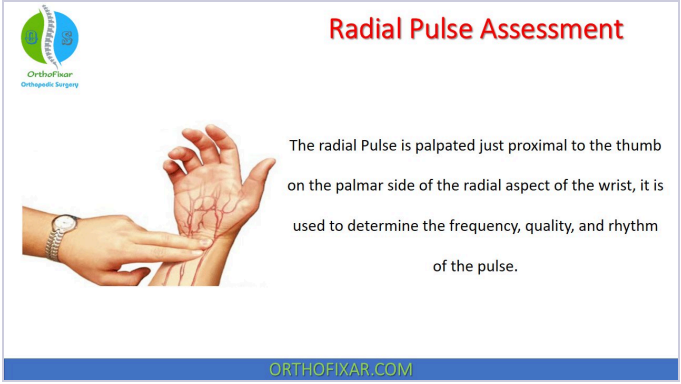
From the FAQ: most interventions are out of your scope, but you can always **recheck and notify the provider (or primary nurse)**. Her non-drug measures (cool bath, cooling blanket, removing extra blankets, oral fluids; warm blankets and fluids for hypothermia) are on Slides 9–10. **NOT IN SLIDES** She doesn't say which of these a first-quarter student may do on their own. Giving an antipyretic is medication administration.

Full 3-part radial pulse assessment [Slides 13–15]

- The pulse is the wave of blood when the **left ventricle contracts**. Sites: carotid, brachial, radial, femoral, popliteal, posterior tibial, dorsalis pedis [13].
- Use your **2nd, 3rd or 4th fingers, never the thumb**. **Count 30 sec × 2**. If irregular, count radial for **1 full minute**, then apical for a full minute. **Apical** = PMI, auscultated for a full minute; peripheral pulses are palpated [14].
- **The 3 parts [15]: (1) Rate**, 60–100, average 80 bpm · **(2) Rhythm**, regular or irregular · **(3) Amplitude**: weak/thready, full/bounding, or imperceptible.
- **Tachycardia = more than 100; bradycardia = less than 60** [Cheat sheet]. Slide 15 leaves these blank.
- FAQ on abnormal pulse: most interventions are out of scope; **recheck the pulse and notify the provider or primary nurse**.



Slide 15 figure: pulse sites (temporal, carotid, apical, brachial, radial, femoral, popliteal, posterior tibial, pedal)



Slide 14: radial pulse assessment, finger pads on the thumb side of the wrist

Electronic vital signs machine

Week 4 lab: the electronic VS machine for **SpO_2, temperature and HR**, plus manual pulse and respirations. TracPrac: pulse oximetry (intermittent or continuous), electronic BP, electronic pulse, temperature (axillary, oral, rectal-discussion), manual pulse, manual respirations, and the documentation exercise [Skills p. 2]. **NOT IN SLIDES** How to operate the machine isn't on her slides. *Steps: see "Not from the professor — double-check" at the end.*

Pulse oximetry [Slide 23]: non-invasive estimate of arterial O_2 saturation (SpO_2). Probes go on the finger, ear, nose or toe. Remove thick nail polish. Wait until the reading is steady and the pulse display is at full strength. With continuous monitoring, check the skin under the probe and rotate it. **SpO_2: normal 95–100%, borderline low 92–94%, hypoxia below 92%** [Cheat sheet].



Slide 23: electronic vital signs machine



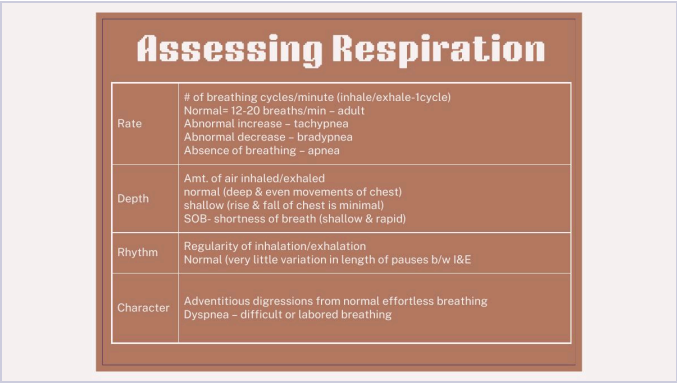
Slide 23: pulse oximetry

Full respiratory rate assessment [Slides 16–17]

- Gas exchange of O_2 and CO_2, under both autonomic and voluntary control. Inspiration = inhalation; expiration = exhalation.
- Count **right after the pulse while still holding the wrist**, so the patient doesn't know you're counting [16].
- **Rate**: one inhale plus one exhale = 1 cycle. **Normal 12–20/min (adult)**. Tachypnea = abnormal increase, bradypnea = abnormal decrease, apnea = no breathing.
- **Depth**: normal (deep, even), shallow (minimal rise), SOB (shallow and rapid). **Rhythm**: regularity, with little variation in pauses. **Character**: **dyspnea** = difficult or labored breathing [17].
- **NOT IN SLIDES** Slide 16's blank "difficulty breathing lying down" term is never filled in on the slides.



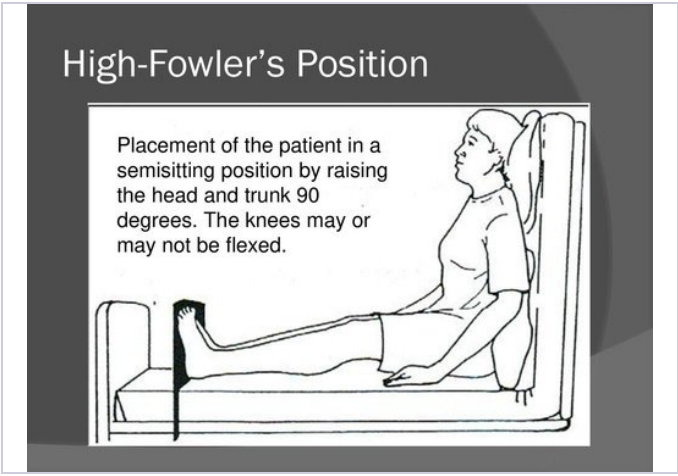
Slide 16: respirations



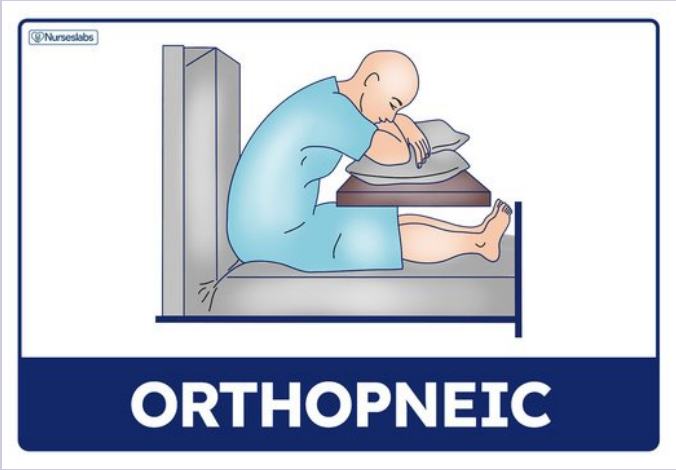
Slide 17: assessing respiration (rate, depth, rhythm, character)

Abnormal RR: positioning, O_2 and scope

From the FAQ: for **SOB or dyspnea, put the patient in high-Fowler's or orthopneic position**. Both positions are pictured on Ch. 34 Slide 12; high-Fowler's = upright at about 90°. **Oxygen administration is NOT within your scope.** Reassess and notify the provider. In her Slide 25 case, the question after a high RR is which other assessment shows oxygenation, i.e. pulse oximetry [Slide 23]. **NOT IN SLIDES** The SOB-to-high-Fowler's rule and the O_2 scope limit come from the FAQ only, not the Vital Signs slides.



Activity Slide 12: high-Fowler's position



Activity Slide 12: orthopneic position

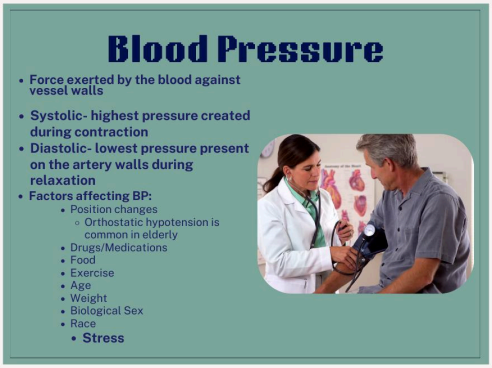
Blood pressure [Slides 19–21]

BP is the force of blood against vessel walls. **Systolic** = highest pressure during contraction; **diastolic** = lowest pressure during relaxation. Factors: position changes (**orthostatic hypotension is common in the elderly**), drugs, food, exercise, age, weight, biological sex, race, stress [19].

Her BP values, exactly as on Slide 20	
Normal	Less than 120/80
Hypotension	Lower than 90 mmHg systolic or 60 mmHg diastolic
Prehypertension (elevated)	Systolic between 120–129 and diastolic of 80
Hypertension Stage 1	130–139 (systolic), 80–89 (diastolic)
Hypertension Stage 2	>140 (systolic), >90 (diastolic)
Hypertensive crisis	>180

> Note: these are her Slide 20 values word for word, and her exam will use them. Her wording overlaps at exactly 80 diastolic, which she lists under both "elevated" and Stage 1, and at 140/90, where Stage 2 says ">140, >90". Her cheat sheet (front page) settles them: elevated = diastolic **<80**, Stage 2 = **140+/90+**, crisis = **over 180 / over 120**.

Technique [21]: cuff size by **arm circumference, not age**; cuff covers **40%** of the arm circumference and the bladder covers **2/3** of the arm; arm supported **at heart level, palm up** (above the heart gives a **false low**). **Cuff too wide = false low; too narrow = false high; too loose = false high.** Korotkoff sounds = sounds of blood flowing through an artery as the cuff is deflated after occluding it. **Limb alerts:** no BP on a limb with a PICC line, an AV fistula, or on the mastectomy side.

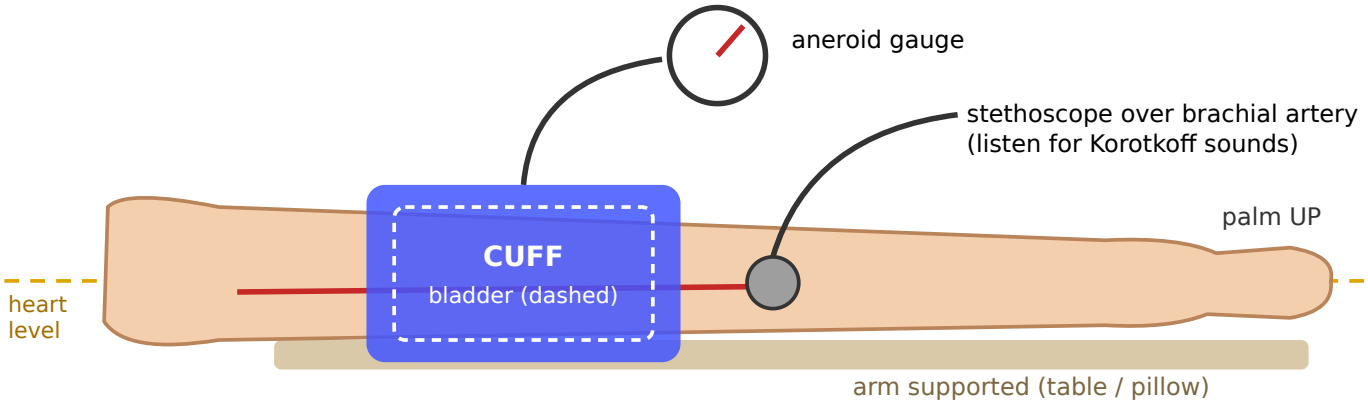


Slide 19: blood pressure



Slide 21: BP assessment with cuff and stethoscope

DIAGRAM · Manual BP cuff placement (her Slide 21 rules)



red line = brachial artery (runs down the inner arm to the antecubital space)

- Choose the cuff by **arm circumference, not age**.
- Cuff covers **40%** of the arm circumference; the bladder covers **2/3** of the arm.
- Arm **at heart level, palm up**. Arm above the heart = **false LOW**.
- Cuff too **wide = false LOW** · too **narrow = false HIGH** · too **loose = false HIGH**.
- No BP on a limb with a PICC line, an AV fistula, or on the mastectomy side.

Diagram (drawn from her Slide 21 rules): cuff and stethoscope placement, arm at heart level

Manual 2-step BP checklist [Weekly Skills Weeks 5–6]

Week 5 lab: **pt 1 = brachial pulse + PSP (palpated systolic pressure)**; **pt 2 = auscultation**. Week 6 = the 2-step manual BP check-off (see the remediation protocol if not passed). Practice on as many adult "patients" as you can. Manual BP goes into TracPrac only after you pass the check-off [Skills p. 3]. **NOT IN SLIDES** The full LLUSN Manual BP checklist steps aren't in these sources; use the checklist and the ATI 2-step video. *Steps: see "Not from the professor — double-check" at the end.*

Vital sign ranges & terminology (from her slides only)

VS	Normal (adult)	Terms
Temp	36–37.5 °C (98.6 °F) [5]; cheat sheet 96.4–100.4 °F	Fever >100.4 °F, hypothermia <95 °F, hyperthermia >104 °F [Cheat sheet]; Febrile/afebrile, pyrexia, hyperpyrexia ≥41 °C (106 °F), hypothermia, hyperthermia

Pulse	60–100, avg 80 [15]	Tachycardia >100 / bradycardia <60 [Cheat sheet], weak/thready, bounding, imperceptible, apical/PMI
RR	12–20 [17]	Tachypnea >20, bradypnea <12 [Cheat sheet], apnea, dyspnea, SOB, shallow
BP	<120/80 [20]	Hypotension, orthostatic hypotension, elevated, Stage 1, Stage 2, crisis, systolic/diastolic, Korotkoff
SpO_2	95–100% [Cheat sheet]	Borderline low 92–94%, hypoxia <92%

What to do when vital signs are abnormal [her FAQ/materials]

ASSESS → VERIFY → RESPOND/REPORT AS APPROPRIATE

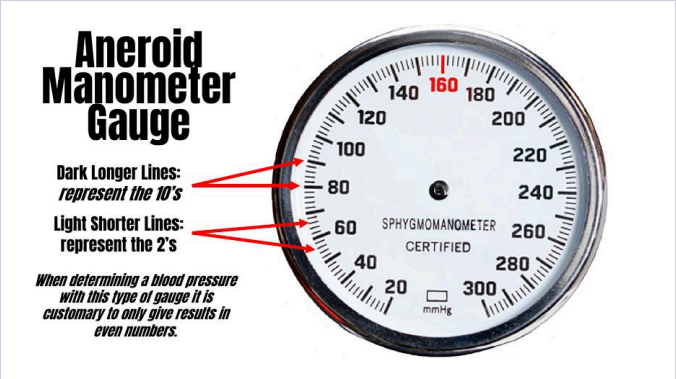
- **Assess:** What do they look like? What am I hearing? What other findings may contribute?
- **Verify** the finding is accurate. An abnormal reading could be from:
  - your technique
  - incorrect equipment or equipment problems
  - incorrect positioning
  - patient factors
  - other circumstances surrounding the measurement
- **Don't automatically panic, but don't ignore it.** Verify, assess, then determine the appropriate response.

Reading an aneroid gauge and BP practice [BP Practice PPT]

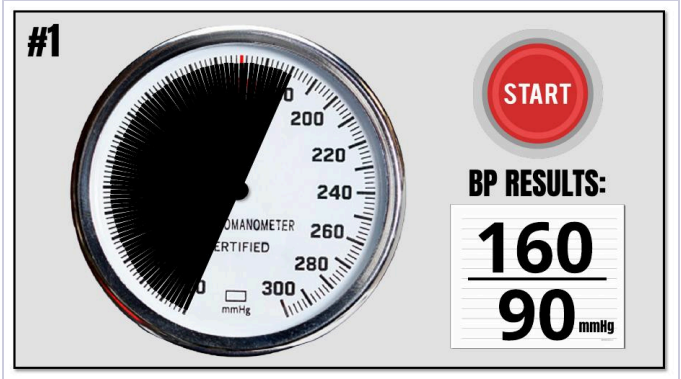
Dark, longer lines = the 10s; light, shorter lines = the 2s. Report BP from this gauge in **even numbers only** [BP PPT Slide 2]. Her gauge-reading practice answers: 130, 96, 52 [Slide 3]. Her BP results, classified with the cheat sheet:

#	Reading	Cheat-sheet category
1	160/90	Stage 2 (140+/90+)
2	130/60	Stage 1 (systolic 130–139)
3	142/84	Stage 2 (systolic 140+)
4	126/68	Elevated (120–129 and <80)
5	168/92	Stage 2

The categories are my classification using her cheat sheet; the PPT shows only the readings. For mixed readings like 142/84, classifying by the higher category is my assumption; her materials don't state the rule.



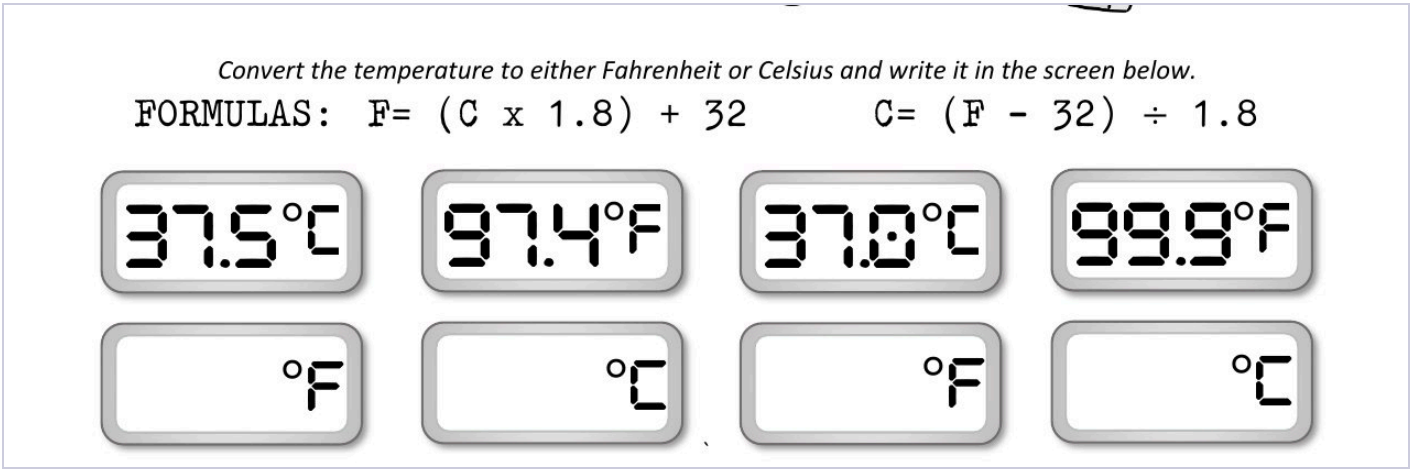
BP PPT Slide 2: aneroid gauge



BP PPT Slide 4: practice #1

Temperature conversion [Extra Practice 3]

Her formulas: **F = (C × 1.8) + 32** and **C = (F − 32) ÷ 1.8**. Her four items, worked: 37.5 °C = **99.5 °F** · 97.4 °F = **36.3 °C** · 37.0 °C = **98.6 °F** · 99.9 °F = **37.7 °C**.



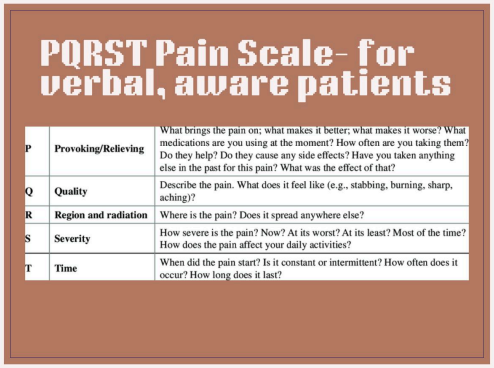
Extra Practice 3 handout: temperature conversion formulas and practice items

PQRST pain assessment [Slide 24], for verbal, aware patients

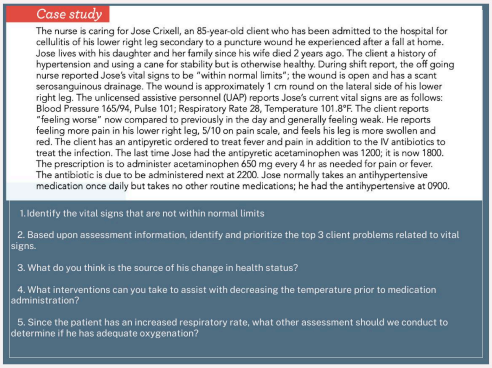
P	Provoking/relieving: what brings it on, what makes it better or worse, current meds and their effect, past treatments
Q	Quality: stabbing, burning, sharp, aching?
R	Region and radiation: where is it, and does it spread?
S	Severity: now, at worst, at least, most of the time; effect on daily activities
T	Time: onset, constant or intermittent, how often, how long

ABC prioritization

The FAQ lists ABC: **airway, breathing, circulation**. **NOT IN SLIDES** No slide teaches ABC. Her Slide 25 case (Jose, 85, cellulitis: BP 165/94, P 101, RR 28, T 101.8 °F, pain 5/10) asks you to prioritize the top 3 VS problems, which is where ABC thinking applies. *Steps: see "Not from the professor — double-check" at the end.*



Slide 24: PQRST



Slide 25: case study

4 · Communication (Part 1) [Ch. 8 slides + Communication Scenario deck]

The communication process & influencing factors [Slides 3, 5–6, 13, 17]

Communication = exchanging information and generating and transmitting meaning between two or more people. It is linked to a healthy work environment (AACN) and to safe, quality care [3]. **Components [5]: sender (encoder)** starts it; **message** (visual, auditory, kinesthetic); **channel** (how it's sent); **receiver (decoder)**; **feedback** (the receiver's response back).

Levels [13]: intrapersonal (self-talk, which can be very damaging), interpersonal (two or more people), group (small group, organizational; group dynamics). **Influencing factors [17]:** developmental level, gender, sociocultural differences, **space and territory** (private zones: head, mouth, feet, face, neck, front of body, genitalia), roles and relationships, occupation/hobbies, physical/mental/emotional state, values, environment, **trust**.

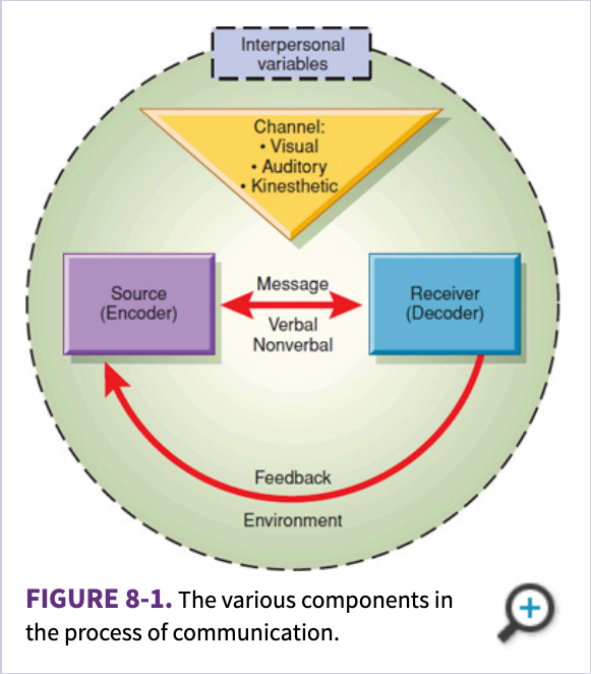


FIGURE 8-1. The various components in the process of communication.

Slide 6: communication process

Nonverbal communication [Slides 7, 11–12]

Verbal = spoken and relies on language. For non-English speakers, an interpreter is the priority. **Nonverbal**: touch, eye contact, facial expression, posture, gait, gestures, appearance/grooming, sounds, silence. **If verbal and nonverbal conflict, believe the nonverbal.** Other forms: electronic (social media, email, text) and paper. Her practice question [12]: stooped gait, grimacing and gasping point to **pain**.



Ch. 8 Slide 11: other forms of communication: electronic (text message)

Interpersonal competencies [Slides 21–22, 24]

Social relationship [21]	Helping (professional) relationship [22]
Mutual, spontaneous, both people's needs considered, no time limit, information shared equally	Meets the patient's needs, has a specific purpose and goals, three phases, a definite end, **unequal sharing of information**

Dispositional traits [24]: warmth/friendliness, openness, respect, empathy, honesty, authenticity, trustworthiness, caring, competence. **Empathy** = seeing things from the patient's point of view. **Sympathy** shifts the focus to the nurse, whose own feelings get projected onto the patient.



Ch. 8 Slide 21: social relationship



Ch. 8 Slide 22: helping (professional) relationship

Phases of the therapeutic (helping) relationship [Slide 23]

Orientation	The patient calls the nurse by name and describes each person's role. Agree on goals, the location, frequency and length of contacts, and how long the relationship lasts.
Working	The patient actively takes part, cooperates toward mutually acceptable goals, and expresses feelings and concerns.
Termination	The patient helps identify goals met or progress made and talks about feelings about ending.

Effective vs ineffective communication [Slides 25, 32–33, 40]

Therapeutic ✓ [Slide 25]	Non-therapeutic ✗ [Slide 33]
Conversation skills (tone, clarity, truthfulness, open mind); active listening (sit, alert, relaxed, eye contact if appropriate); **silence**; touch (shoulder or hand, otherwise ask consent); humor used carefully. Questions: **open-ended** ("How are you feeling right now?"), **validating** ("You seem upset, can we talk about it?"), **clarifying** ("Did I understand you correctly…?"), **reflective** ("What went well today?"). Focus on feelings; **state observed behaviors** ("I can see you have been crying…"); reflect/restate/rephrase ("It sounds like you're frustrated…"); neutral responses; **exploring** ("Let's talk about it").	Closed-ended questions; forgetting they are human; failure to listen; **leading questions** ("You're not really sad, right?"); **giving advice**; **changing the subject**; **talking about yourself** ("My grandma also had cancer"); telling the patient they're wrong; **false reassurance**; **clichés** ("Everything happens for a reason"); **giving approval** ("I think you are making the right choice"); **asking why**; **defending other staff** ("Sally is actually a fantastic nurse…").

Her practice questions: the patient is silent after hearing a diagnosis, so **allow time to think** [32]. "I am sure everything will be fine" = **false reassurance** [40].

Role-play scenarios: technique vs barrier [Communication Scenario deck]

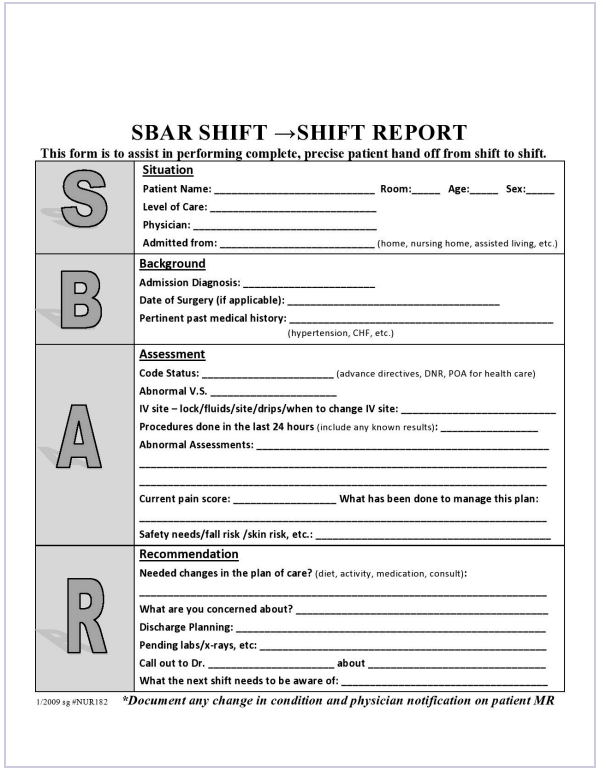
#	Patient	Therapeutic	Non-therapeutic
1	Angry, long ED wait	Reflective question/comment	Aggressive behavior
2	Nervous, IV start	Gentle, appropriate humor	Arguing
3	Hopelessness	Exploring statement (assess safety)	Failure to listen
4	Doesn't understand plan	Open-ended questions	Accusatory "why" questions
5	Anxious pre-op	Realistic reassurance and support	False reassurance/promises
6	Frustrated, chronic HTN	Active listening	Judgmental comments
7	Grieving family member	Emotional/spiritual support	Clichés
8	Confused elderly	Presenting reality, reorientation	Minimizing concerns
9	Isolated adolescent	Restating/mirroring	Nonverbal boredom
10	Refusing treatment	Clarifying statement	Disapproving comments
11	Overwhelmed new parent	Validating statement	Giving advice
12	Unexpected post-op pain	Direct, focused question	Leading question
13	Teen, body image	Empathetic statement	Intrusive/premature probing
14	Seriously ill, afraid of dying	Therapeutic touch, with permission	Talking about yourself
15	Behavior doesn't match pain	Making observations	Ignoring nonverbal cues
16	Transgender, respect	Affirming identity	Dehumanizing/dismissive
17	Parent of chronically ill child	Therapeutic silence	Only yes/no questions
18	Student near burnout	Collaborative problem-solving	Gossip/rumors
19	Parent learning diabetes care	Offering self and support	Poor professional presentation
20	Limited English	Qualified medical interpreter	Frustrated/dismissive

Interventions for impaired communication [Slides 41–43]

- **Visually impaired**: announce yourself and give your name; speak in a normal tone; explain before touching; say when you're leaving; call light within reach; orient them to room sounds and layout; glasses clean, contacts in.

- **Hearing impaired**: get their attention first (gentle touch or move into view); face them; simple sentences at a natural pace for lip reading; no gum or covering your mouth; pantomime, sign or write; hearing aids clean, working and in place.
- **Physical barrier (laryngectomy, ET tube)**: simple means such as eye blinks or hand squeezes for yes/no, writing pads, communication boards, flash cards, sign language; make sure everyone can use the method; be patient; make sure they can signal for help (call bell).
- **Cognitively impaired**: eye contact; quiet setting; simple, concrete steps without pronouns or abstract terms; **avoid open-ended questions and offer choices** ("brown pants or gray?"); if there's no response after **2 minutes**, repeat, then take a break.
- **Unconscious**: hearing is the **last sense lost**, so assume they can hear; talk normally; speak before touching; keep noise low.
- **Non-English speaking**: **use an interpreter whenever possible**; a translation dictionary; simple sentences in a normal tone; pantomime; watch nonverbal cues.

SBAR [Slide 44]: Situation, Background, Assessment, Recommendation. Used for hand-off and nurse-to-provider communication to make communication more effective.



SBAR SHIFT →SHIFT REPORT

This form is to assist in performing complete, precise patient hand off from shift to shift.

S — Situation

Patient Name: _____ Room:_____ Age:_____ Sex:_____
Level of Care: _____
Physician: _____
Admitted from: _____ (home, nursing home, assisted living, etc.)

B — Background

Admission Diagnosis: _____
Date of Surgery (if applicable): _____
Pertinent past medical history: _____
(hypertension, CHF, etc.)

A — Assessment

Code Status: _____ (advance directives, DNR, POA for health care)
Abnormal V.S. _____
IV site – lock/fluids/site/drips/when to change IV site: _____
Procedures done in the last 24 hours (include any known results): _____
Abnormal Assessments: _____
_____
_____
Current pain score: _____ What has been done to manage this plan:
_____
Safety needs/fall risk /skin risk, etc.: _____

R — Recommendation

Needed changes in the plan of care? (diet, activity, medication, consult):
_____
What are you concerned about? _____
Discharge Planning: _____
Pending labs/x-rays, etc: _____
Call out to Dr. _____ about _____
What the next shift needs to be aware of: _____

1/2009 sg #NUR182 **Document any change in condition and physician notification on patient MR*

Ch. 8 Slide 45: SBAR shift report form

Her Week 4 Wrap Up announcement says the Communication PPT is midterm material even though it wasn't covered in class.

Medical Terminology (Ehrlich)

Source: Ehrlich et al., *Medical Terminology for Health Professions*, 8th ed. and the Med Term course handout ("Handout"). **Not the Foundations professor's material, but confirmed on the midterm.** Chapters = what Med Term Quizzes #1 (Word Parts & Body Directions) and #2 (Immunity & Blood) covered. Page numbers are Ehrlich's printed pages. ↔ Foundations rows (highlighted) overlap her vital signs, infection, positioning or communication content.

Ch. 1 · Introduction to Medical Terminology (Quiz #1: word parts)

Ch. 1 · Suffixes

Word part / term	Meaning	Ehrlich
-algia / -dynia ↔ Foundations	pain, suffering	p. 2, 6
-itis ↔ Foundations	inflammation	p. 2, 6
-osis	abnormal condition, disease	p. 2, 7
-megaly	enlargement	p. 7
-malacia	abnormal softening	p. 7
-sclerosis	abnormal hardening	p. 2
-necrosis	tissue death	p. 7–8
-stenosis	abnormal narrowing	p. 7–8
-oma	tumor	p. 6
-cyte	cell	p. 5
-ectomy	surgical removal, cutting out	p. 2
-otomy	cutting, surgical incision	p. 2
-ostomy	surgical creation of an artificial opening	p. 2
-plasty	surgical repair	p. 2
-rrhaphy	surgical suturing	p. 2
-centesis	surgical puncture to remove fluid	p. 7
-graphy / -gram	process of recording a picture / the picture itself	p. 7
-scopy	visual examination	p. 7
-rrhage	bleeding; abnormal excessive fluid discharge	p. 2
-rrhea	flow or discharge	p. 2, 8
-rrhexis	rupture	p. 2
-ac, -al, -ic, -ous, -tic …	pertaining to	Table 1.4

Ch. 1 · Prefixes

Word part / term	Meaning	Ehrlich
a-, an-	without, not	Handout
ab- ↔ Foundations	away from	Table 1.6–1.7
ad- ↔ Foundations	toward	Table 1.6–1.7
dys- ↔ Foundations	bad, difficult, painful	p. 2, Table 1.7
eu- ↔ Foundations	good, normal, easy	Table 1.7
hyper- ↔ Foundations	excessive, increased	Table 1.7

hypo- ↔ Foundations	deficient, decreased	Table 1.7
inter-	between, among	Table 1.7
intra-	within, inside	Table 1.7
sub-	under, below	Table 1.7
super-, supra-	above, excessive	Table 1.7
ex-	out of, outside	Table 1.6
in-	in, into, not	Table 1.6
macro-	large	Table 1.6
micr/o-	small	Table 1.6
olig/o	scanty, few	Table 1.6
pre- / post-	before / after	Table 1.6
dextr/o / sinistr/o	right / left side	Table 1.6

Ch. 1 · Combining forms (color + handout roots)

Word part / term	Meaning	Ehrlich
cyan/o ↔ Foundations	blue	Table 1.2
erythr/o	red	Table 1.2
leuk/o ↔ Foundations	white	Table 1.2
melan/o	black	Table 1.2
poli/o	gray	Table 1.2
derm/o, dermat/o; cutane/o	skin	Handout
nephr/o; ren/o	kidney	Handout
stomat/o; or/o ↔ Foundations	mouth	Handout
gastr/o	stomach	p. 6
hepat/o	liver	p. 7
oste/o	bone	Handout
immun/o ↔ Foundations	immune	Handout
myel/o	spinal cord (also bone marrow)	Handout; p. 5
neur/o	nerve	Handout
crani/o	skull	Handout
cardi/o ↔ Foundations	heart	Handout
my/o	muscle	Handout
pyel/o	renal pelvis	Handout
arthr/o	joint	p. 4

Ch. 1 · Key terms built from them

Word part / term	Meaning	Ehrlich
cyanosis ↔ Foundations	blue skin from lack of oxygen	p. 5
erythrocyte / leukocyte ↔ Foundations	red / white blood cell	p. 5
gastritis; gastralgia	stomach inflammation; stomach pain	p. 6
hepatomegaly	enlarged liver	p. 7
arteriostenosis	narrowed artery	p. 7–8
abdominocentesis	puncture of abdomen to remove fluid	p. 7
hypertension / hypotension ↔ Foundations	higher / lower than normal BP	Table 1.7

eupnea ↔ Foundations	easy, normal breathing	Table 1.7
dysfunctional	not working properly	Table 1.7
intramuscular	within the muscle	Table 1.7
subcostal / supracostal	below / above the ribs	Table 1.7
interstitial	between the parts of a tissue	Table 1.7
diagnosis → diagnoses	-is plural becomes -es	Table 1.9

Ch. 2 · The Human Body in Health and Disease (Quiz #1: body directions)

Ch. 2 · Word parts

Word part / term	Meaning	Ehrlich
aden/o	gland	p. 29
adip/o	fat	p. 29
anter/o ↔ Foundations	before, front	p. 29
poster/o ↔ Foundations	behind, back	p. 29
cephal/o	head	p. 29
caud/o	lower body, tail	p. 29
cyt/o, -cyte	cell	p. 29
end-, endo-	in, within	p. 29
exo-	out of, outside	p. 29
hist/o	tissue	p. 29
path/o, -pathy ↔ Foundations	disease, suffering	p. 29
plas/i, -plasia	development, growth	p. 29
-stasis ↔ Foundations	control, keep a constant level	p. 29
-ology / -ologist	study of / specialist	p. 29

Ch. 2 · Body planes & directions (Quiz #1 “Body Directions”)

Word part / term	Meaning	Ehrlich
midsagittal plane	divides body into equal left and right halves (midline)	p. 31
frontal (coronal) plane	divides front (anterior) from back (posterior)	p. 31
transverse plane	divides upper (superior) from lower (inferior)	p. 31
superior / inferior ↔ Foundations	above, toward head / below, toward feet	p. 32
cephalic / caudal	toward the head / toward the feet	p. 32
anterior (ventral) / posterior (dorsal) ↔ Foundations	front / back	p. 33
medial / lateral ↔ Foundations	toward midline / toward the side	p. 33
bilateral	both sides	p. 33
proximal / distal ↔ Foundations	nearest / farthest from midline or attachment	p. 33
superficial / deep	near the surface / away from the surface	Handout list; Ch. 4
abduction / adduction ↔ Foundations	move limb away from / toward the midline	Ch. 4 (movement)
inversion / eversion	turning inward / outward	Handout list

Ch. 2 · Disease terms

Word part / term	Meaning	Ehrlich
homeostasis ↔ Foundations	keeping a stable internal environment	p. 29
etiology	study of the cause of disease	p. 29
idiopathic	disease of unknown cause	p. 29
iatrogenic	illness caused by treatment	p. 29
nosocomial infection ↔ Foundations	infection acquired in a hospital (HAI)	p. 29
communicable disease ↔ Foundations	spread person to person	p. 29
bloodborne / vector-borne ↔ Foundations	spread by blood / by a vector (e.g., mosquito)	p. 29
endemic / epidemic / pandemic ↔ Foundations	constant in an area / sudden spread / worldwide	p. 29
congenital disorder	present at birth	p. 29
hypertrophy / hyperplasia	larger cells / more cells	p. 29
aplasia / hypoplasia / dysplasia	no / incomplete / abnormal development	p. 29

Ch. 2 · Abbreviations

Word part / term	Meaning	Ehrlich
A & P	anatomy and physiology	Table 2.2
CD ↔ Foundations	communicable disease	Table 2.2
DNA	deoxyribonucleic acid	Table 2.2
epid	epidemic	Table 2.2
GP / PA	general practitioner / physician assistant	Table 2.2
RUQ RLQ LUQ LLQ	right/left upper/lower quadrant	Table 2.2

Ch. 5 · Cardiovascular, blood portion only (Quiz #2: blood)

Ch. 5 (blood portion) · Word parts

Word part / term	Meaning	Ehrlich
hem/o, hemat/o	blood	p. 123
-emia	blood condition	p. 123
erythr/o	red	p. 123
leuk/o	white	p. 123
thromb/o	clot	p. 123
-crasia	mixture or blending	p. 123
brady- / tachy- ↔ Foundations	slow / fast	p. 123
angi/o	blood or lymph vessel	p. 123
phleb/o, ven/o	vein	p. 123
arteri/o	artery	p. 123

Ch. 5 (blood) · Key terms

Word part / term	Meaning	Ehrlich
plasma / serum	liquid part of blood / plasma without cells and clotting proteins	p. 133
erythrocytes (RBCs)	red blood cells; carry oxygen via hemoglobin	p. 133
hemoglobin ↔ Foundations	oxygen-carrying protein in RBCs	p. 133
leukocytes (WBCs) ↔ Foundations	white cells that fight infection	p. 133

neutrophils / eosinophils / basophils ↔ Foundations	most common WBC (phagocytosis) / parasites & allergy / allergy symptoms	p. 133
lymphocytes / monocytes	identify germs / phagocytosis	p. 133
thrombocytes (platelets)	help blood clot	p. 133
phagocytosis ↔ Foundations	engulfing and destroying pathogens	p. 133
antigen ↔ Foundations	anything the body sees as foreign	p. 134
blood types; Rh factor	A, B, AB, O; Rh antigen present (+) or not (−)	p. 134
hemolysis	breakdown of RBCs	p. 144
hematologist	blood specialist	p. 134
bradycardia / tachycardia ↔ Foundations	slow / fast heart rate	p. 123 parts

Ch. 6 · Lymphatic & Immune Systems (Quiz #2: immunity)

Ch. 6 · Word parts

Word part / term	Meaning	Ehrlich
immun/o ↔ Foundations	immune, protection	p. 164
lymph/o	lymph	p. 164
lymphaden/o	lymph node	p. 164
lymphangi/o	lymph vessel	p. 164
splen/o	spleen	p. 164
anti- ↔ Foundations	against	p. 164
phag/o ↔ Foundations	eat, swallow	p. 164
tox/o	poison	p. 164
onc/o; -oma	tumor	p. 164
carcin/o	cancerous	p. 164
sarc/o	flesh, connective tissue	p. 164
neo-	new	p. 164
-plasm	formative cell material	p. 164

Ch. 6 · Key terms

Word part / term	Meaning	Ehrlich
antibody / immunoglobulin ↔ Foundations	protein made against a specific antigen	p. 171
B cells / T cells	make antibodies / cell-mediated immunity (mature in thymus)	p. 168
natural killer cells	kill cancer and virus-infected cells	p. 168
macrophage ↔ Foundations	swallows invaders and dead cells	p. 172
immunity ↔ Foundations	resistance to a specific disease	p. 172
natural (passive) vs acquired immunity ↔ Foundations	present at birth vs from disease or vaccination	p. 172
allergen; histamine	causes allergy; causes itching/sneezing	p. 174
lymph nodes (cervical, axillary, inguinal) ↔ Foundations	bean-shaped filters with lymphocytes; neck / armpit / groin	p. 166
spleen; thymus	filters blood / matures T cells	p. 169
pathogen; pathogenic ↔ Foundations	disease-causing germ; able to cause disease	p. 177

bacteria, virus, fungus, parasite ↔ Foundations	the 4 kinds of infectious agents	p. 177–179
staphylococci / streptococci ↔ Foundations	grape-like clusters / chains	p. 178
Staphylococcus aureus ↔ Foundations	infects wounds; toxic shock, food poisoning	p. 178
nosocomial ↔ Foundations	hospital acquired	p. 178

Ch. 6 · Abbreviations

Word part / term	Meaning	Ehrlich
Ab / Ag ↔ Foundations	antibody / antigen	Table 6.3
Ig	immunoglobulin	Table 6.3
CA, Ca	cancer	Table 6.3
CIS	carcinoma in situ	Table 6.3
HL / NHL	Hodgkin's / non-Hodgkin's lymphoma	Table 6.3
HZ	herpes zoster	Table 6.3
MMR	measles, mumps, rubella vaccine	Table 6.3
MET / met	metastasis / metastasize	Table 6.3

Midterm Vocab (Modules 1–4)

Short definitions from her slides and her Vital Signs Cheat Sheet only. Source after each term (S = slide). Terms her materials name but don't define are at the very end under **Not from the professor — double-check**. Modules follow her Exam FAQ.

Module 1 · Ch. 25 Asepsis & Infection Control

Term	Meaning	Source
Infection	A disease state caused by pathogens in or on the body. Needs all six links of the chain to happen.	Asepsis S3
Chain of infection	Infectious agent → reservoir → portal of exit → means of transmission → portal of entry → susceptible host. Break any link to stop infection.	Asepsis S3–4
Infectious agent	The germ: bacteria (E. coli, staph, strep), virus (HIV, flu, COVID-19, hepatitis), fungi (athlete's foot, yeast), parasites (malaria).	Asepsis S5
Reservoir	Where the germ lives: people, animals, soil, food, water, milk, objects.	Asepsis S6
Portal of exit / entry	How germs leave or get in: respiratory, GI, GU tracts, skin breakdown, blood (and drainage/tissues).	Asepsis S7, S11
Direct contact	Spread by close contact: touching, kissing, sex.	Asepsis S8
Indirect contact	Spread through a vector or a fomite.	Asepsis S8
Vector	A living creature that carries a pathogen to humans (e.g., mosquito).	Asepsis S8
Fomite	A non-living object that holds a pathogen (e.g., a stethoscope).	Asepsis S8, S10
Droplet transmission	Particles bigger than 5 µm; heavy enough to land on surfaces.	Asepsis S9
Airborne transmission	Particles smaller than 5 µm; stay in the air.	Asepsis S9
Susceptible host	Factors that raise risk: wounds/poor skin, low WBC, age, immunity, fatigue, poor nutrition, existing illness, some meds, stress, invasive/indwelling devices.	Asepsis S12
Convalescent stage	Illness resolved; recovery and return to baseline health.	Asepsis S17
Stages of infection	Incubation → prodromal (hours to days) → full stage of illness → convalescent.	Asepsis S17
Medical asepsis	“Clean technique.” Reduces the number of pathogens (hand hygiene, gloves, clean to dirty).	Asepsis S19
Surgical asepsis	“Sterile technique.” Eliminates pathogens; a pathogen-free field (Foley, central lines).	Asepsis S19
Five Moments for Hand Hygiene	Before touching a patient; before a clean/aseptic procedure; after body-fluid risk; after touching a patient; after touching patient surroundings.	Asepsis S21
HAI (healthcare-associated infection)	Also called hospital-acquired or nosocomial. Main ones: CAUTI (catheter UTI), SSI (surgical site), CLABSI (central-line bloodstream), VAP (ventilator pneumonia).	Asepsis S22
MDRO (multidrug-resistant organism)	VRE (feces, urine, blood), MRSA (nares, skin, GI), CRE, C. diff (feces).	Asepsis S23, S29
PPE	Gloves, gown, masks, protective eyewear.	Asepsis S25
Standard precautions	Used for ALL patients. Covers blood, all body fluids except sweat, non-intact skin, mucous membranes.	Asepsis S26
Transmission-based precautions	Added on top of standard precautions for airborne, droplet or contact germs. Don PPE entering, doff leaving.	Asepsis S27
Droplet precautions	Private room, door can stay open, surgical mask + gloves.	Asepsis S28

Contact precautions	For MDROs. Private room, gown + gloves, clean equipment and don't share it.	Asepsis S29
Airborne precautions	Private room, negative air pressure, door closed, respirator mask; surgical mask on patient for transport.	Asepsis S30
Reverse / neutropenic precautions	Protect severely immunocompromised patients (cancer treatment, transplant, WBC below 500).	Asepsis S31

Module 2 · Ch. 34 Safe Patient Handling (Activity lecture)

Term	Meaning	Source
Body alignment (posture)	Best musculoskeletal balance with no extra strain on joints, muscles, tendons or ligaments.	Activity S3
Balance	Center of gravity close to a wide base of support. Spread feet and bend hips/knees to improve it.	Activity S3, S7
Coordinated body movement	Muscles working together for purposeful movement, using the big, strong muscle groups.	Activity S4
Postural reflexes	Keep body position and balance at rest or moving (inner ear, vision, autonomic contractions).	Activity S4
ADLs (activities of daily living)	Self-care activities that keep quality of life: eating, bathing, dressing, toileting.	Activity S5
Ergonomics	Designing equipment and tasks to fit the worker's ability, to prevent injury.	Activity S6
Body mechanics rules	Use legs/arms, not back; keep object close; face the way you move, don't twist; push rather than pull; feet shoulder-width, knees flexed.	Activity S7
Mobility devices	Friction-reducing sheets, standing assist, gait belts, lateral assist device, powered full-body lifts.	Activity S11
Fowler's / semi- / high-Fowler's	Head of bed 45–60° / 15–45° / about 90°.	Activity S12 (positions picture)

Module 3 · Vital Signs pt 1 & 2 (Ch. 26 slides + cheat sheet)

Term	Meaning	Source
Vital signs	Show how the body is working: temp, pulse, respirations, BP, pulse ox, pain.	VS S3
Thermoregulation	The hypothalamus controls body heat by saving heat or increasing heat loss.	VS S5
Core temperature	Inside temp (intracranial, intrathoracic, intra-abdominal). Cells work best 36–37.5 °C.	VS S5
Circadian rhythm	Predictable daily temp changes: lower in the morning, peak 1500–1800.	VS S6
Febrile / afebrile	Has a fever / no fever.	VS S7
Pyrexia (fever)	Over 100.4 °F (38 °C).	Cheat sheet; VS S7
Hyperthermia	Over 104 °F (40 °C).	Cheat sheet
Hyperpyrexia	106 °F (41 °C) or more. Medical emergency.	Cheat sheet; VS S7
Hypothermia	Below 95 °F (35 °C). Signs: poor coordination, slurred speech, confusion; RR, HR, BP drop.	Cheat sheet; VS S10
Normal temperature	96.4–100.4 °F.	Cheat sheet
Oral / sublingual site	Posterior sublingual pocket under tongue. No hot/cold drinks or smoking 20 min before. 3 min.	VS S12

Axillary	Middle of armpit. Least accurate; 0.5 °C lower than oral. 5–10 min.	VS S12
Rectal	Side-lying, upper leg flexed. 0.5 °C higher than oral. 2–3 min.	VS S12
Tympanic (ear)	Near the hypothalamus. Adult: pull pinna up and back. 2–3 sec. Earwax or ear infection can distort it.	VS S11–12
Pulse	Wave of blood when the left ventricle contracts; felt over an artery against bone.	VS S13
Pulse sites	Carotid, brachial, radial, femoral, popliteal, posterior tibial, dorsalis pedis.	VS S13
Peripheral pulse	Palpate with 2nd–4th fingers, never thumb. 30 sec ×2; if irregular, radial 1 min then apical 1 min.	VS S14
Apical pulse / PMI	Heartbeat at the heart's apex; auscultate a full minute.	VS S14
Pulse rhythm	Pattern of beats: regular or irregular.	VS S15
Pulse amplitude	Volume of blood pushed against the artery wall: thready (weak), bounding (full), imperceptible (can't be felt).	VS S15
Bradycardia / tachycardia	HR under 60 / over 100. Normal 60–100.	Cheat sheet; VS S15
Inspiration / expiration	Breathing in / breathing out.	VS S16
Bradypnea / tachypnea	RR under 12 / over 20. Normal 12–20.	Cheat sheet; VS S17
Apnea	No breathing.	VS S17
Dyspnea	Difficult or labored breathing.	VS S17
SOB	Shortness of breath: shallow and rapid.	VS S17
Blood pressure	Force of blood against vessel walls.	VS S19
Systolic / diastolic	Highest pressure during contraction / lowest during relaxation.	VS S19
BP categories	Normal <120/80; elevated 120–129/<80; stage 1 130–139/80–89; stage 2 140+/90+; crisis >180/>120.	Cheat sheet
Hypotension	Below 90 systolic or below 60 diastolic.	Cheat sheet; VS S20
Korotkoff sounds	Sounds of blood flowing through an artery as the cuff pressure is let down.	VS S21
Cuff errors	Cuff 40% of arm circumference, bladder 2/3 of arm. Too wide or arm above heart = false low; too narrow or loose = false high.	VS S21
Limb alerts	No BP on an arm with a PICC line, AV fistula, or mastectomy side.	VS S21
Pulse oximetry (SpO_2)	Non-invasive estimate of arterial oxygen saturation. Normal 95–100%.	VS S23; Cheat sheet
Hypoxia	SpO_2 below 92% (92–94% borderline low).	Cheat sheet
Baseline vital signs	A first set of VS used to spot later changes.	VS S26

Module 4 · Communication pt 1 (Ch. 8 slides)

Term	Meaning	Source
Communication	Exchanging information and creating/sending meaning between two or more people.	Comm S3
Sender (encoder)	Starts the message.	Comm S5
Message	The actual communication (visual, auditory, kinesthetic).	Comm S5
Channel	How the message is sent (audio, visual, kinesthetic).	Comm S5
Receiver (decoder)	Person the message is sent to.	Comm S5
Feedback	The receiver's response back to the sender.	Comm S5

Verbal / nonverbal	Spoken language / body language (touch, eye contact, face, posture, gait, gestures, appearance, silence).	Comm S7
Intrapersonal / interpersonal	Self-talk / between two or more people.	Comm S13
Space and territory	Private zones: head, mouth, feet, face, neck, front of body, genitals.	Comm S17
Social relationship	Spontaneous, mutual, no time limit, equal sharing.	Comm S21
Helping (professional) relationship	Goal-directed, three phases, a definite end, unequal sharing.	Comm S22
Empathy	Seeing things from the patient's point of view.	Comm S24
Sympathy	Shifts focus to the nurse; the nurse's feelings get projected onto the patient.	Comm S24
Active listening	Sit, alert, relaxed, natural, eye contact if appropriate.	Comm S25
Therapeutic silence	Lets the patient speak freely / think.	Comm S25, S32
Open-ended question	Invites a full answer: “How are you feeling right now?”	Comm S25
Validating / clarifying / reflective	Check (“You seem upset…”) / ask to explain (“Did I understand you correctly…”) / look back (“What went well today?”).	Comm S25
Restating / reflecting	Repeat or rephrase what the patient said: “It sounds like you're frustrated…”	Comm S25
Leading question	Pushes an answer: “You're not really sad, right?”	Comm S33
False reassurance	“I'm sure everything will be fine.”	Comm S33, S40
Cliché	“Everything happens for a reason.”	Comm S33
Giving approval	“I think you are making the right choice.”	Comm S33
Defending	Defending staff: “Sally is actually a fantastic nurse…”	Comm S33
SBAR	Situation, Background, Assessment, Recommendation. Used for hand-off and nurse-to-provider talks.	Comm S44

Gaps: FAQ items her materials don't answer (answers in the last section, “Not from the professor — double-check”)

- Hand hygiene times and when sanitizer isn't appropriate (Slide 24 asks without answering)
- PPE donning/doffing order
- HAI prevention specifics beyond hand hygiene, asepsis for Foleys and central lines, and device risk
- Step-by-step: ambulation with safe fall, wheelchair transfer, lateral transfer to gurney, repositioning in bed
- Temporal technique steps (the handout only says "forehead"); older-adult and women temperature questions (Slide 6); which VS rises with fever (Slide 8); answer key for Extra Practice 3 items 6–10
- Extra Practice 1 blanks with no answers in her materials: Cheyne-Stokes, rales, VS documentation order, the 10-second pulse example
- First-quarter scope for abnormal temperature (FAQ: recheck and notify)
- The "difficulty breathing lying down" term (blank on Slide 16)
- Electronic VS machine operation
- High-Fowler's/orthopneic for SOB and O_2 out of scope (FAQ only)
- LLUSN manual 2-step BP checklist steps
- ABC prioritization

Not from the professor — double-check

Everything below answers FAQ items her slides, handouts, skills sheets and cheat sheet do not answer. Standard nursing sources, not hers. Check against her lab checklists or ask her.

A. Standard answers to FAQ gaps

> These answers are **not from her materials**. They come from standard references (CDC, WHO, AHA, NIOSH, the Open RN *Nursing Skills* open textbook and device manuals), and **none are from Taylor**. If her ATI or LLUSN checklist says something different, go with her checklist.

1. Hand hygiene: times and when sanitizer is not appropriate [Slide 24]

- **Alcohol rub:** a palmful covering all surfaces, rub **~20 sec until dry**. This is the preferred method for most clinical care because it kills more germs and is faster.
- **Soap & water:** wet hands, lather and scrub **at least 20 sec** (all surfaces, between fingers, thumbs, under nails), rinse, dry with a towel, and turn off the faucet with the towel.
- **Use soap & water, NOT sanitizer, when:** hands are **visibly soiled**; caring for a patient with **C. diff** or **norovirus** (alcohol doesn't kill C. diff spores); before eating; after using the restroom.
- **Best way to prevent infection [Slide 32]:** hand hygiene.

Source: CDC, Hand Hygiene in Healthcare Settings; WHO Five Moments.

2. PPE donning & doffing order

Don (outside the room)	Doff (at the door / anteroom)
1 Hand hygiene → 2 **Gown** (tie at neck and waist) → 3 **Mask or respirator** (mold the nose piece; fit-check an N95) → 4 **Goggles / face shield** → 5 **Gloves** (pulled over the gown cuffs)	1 **Gloves** (most contaminated; glove-to-glove, then skin-to-skin) → 2 **Goggles / face shield** (handle by the band) → 3 **Gown** (pull away from the body and roll it inside-out) → 4 **Mask/respirator** (by the ties or straps; take off an N95 **after leaving** the room) → 5 **Hand hygiene**

- Do hand hygiene right away if your hands get contaminated at any step. CDC also accepts removing the gown and gloves together.

Source: CDC, "Sequence for Putting On / Removing PPE" (CS250672).

3. HAI prevention specifics [Slides 22, 24]

- **All HAIs:** hand hygiene, standard/transmission precautions, clean equipment, and remove devices as soon as they aren't needed.
- **CAUTI:** insert only when indicated, using sterile technique; keep a closed system; keep the bag **below the bladder and off the floor**; no kinks; do daily perineal care; review the need daily and remove early.
- **CLABSI:** sterile insertion with chlorhexidine (CHG) skin prep; **"scrub the hub"** before every access; keep the dressing clean, dry and intact, and change it per policy; review the need daily.
- **SSI:** pre-op CHG bathing; clip hair, don't shave; keep the dressing sterile; monitor glucose; watch for redness, drainage and fever.
- **VAP: HOB 30–45°**, frequent oral care, suctioning with good technique; daily sedation and weaning checks are done by the provider.
- **Slide 24, "patient with a device shows infection signs":** suspect the device as the source (an HAI). Report it, and the device may need to be removed or cultured.

Source: CDC/HICPAC guidelines for CAUTI, intravascular catheter infections, SSI; IHI ventilator bundle.

4. Ch. 34 lab skills: step by step

Before every move: hand hygiene, two identifiers, explain the move, enough staff, bed locked at a working height, non-skid footwear, and a gait belt if the patient can bear weight.

- **Ambulation with a safe (assisted) fall:** dangle the patient on the edge of the bed and check for dizziness. Walk on the **weak side, slightly behind**, with an underhand grip on the gait belt. If the patient starts to fall: **don't try to hold them up**. Widen your stance, pull them close against your hip, and **slide them down your leg to the floor** while protecting their head. Stay with them, call for help, assess them before moving them, and report it.
- **Bed → wheelchair (stand-pivot):** put the chair on the patient's **strong side at about 45°** to the bed, **lock the wheels** and swing the footrests away. Lower the bed. The patient sits at the edge with feet flat. Face the patient, hold the gait belt, bend your knees and brace their knees. "On 3" they stand, **pivot** toward the strong side, and sit back once the chair touches the backs of their legs. Then put the footrests back, position them and leave the call light in reach.
- **Lateral transfer bed → gurney:** use **at least 2–3 staff** and a **friction-reducing sheet or slide board**. Put the bed flat, line the gurney up level with the bed (or slightly lower) and **lock both**. Logroll the patient to place the device. "On 3," **pull rather than push or reach** while keeping your back straight. Then raise the rails and secure the straps.
- **Repositioning up in bed:** lay the bed flat at **waist height**, lock it, and use 2 staff with a friction-reducing/draw sheet. The patient bends their knees, tucks their chin and crosses their arms. Use a wide stance, and "on 3" shift your weight from your back leg to your front leg. Turning/side-lying: logroll, then use pillows behind the back and between the knees. Reposition at least **every 2 h**.

Source: NIOSH/ANA Safe Patient Handling; Open RN *Nursing Skills* (2nd ed.), Ch. 3 Mobility.

5. Temperature questions she leaves open

- **Temporal technique:** forehead clean and dry, hair pushed back. Hold the button and **slide straight across the forehead, midline to the hairline**. If the patient is sweating, keep holding and touch behind the earlobe. Release the button and read. (Exergen instructions)
- **Older adults [Slide 6]:** run **lower** (often about 96.8 °F / 36 °C) and have a weaker fever response. A "normal" temp can still be an infection.
- **Women [Slide 6]:** temperature changes with hormones. It's about **0.3–0.6 °C higher after ovulation** (progesterone), and menopause can bring hot flashes.
- **Which VS rises with fever [Slide 8]: pulse**, about 10 bpm for every 1 °C, and **respirations** (higher metabolic rate).
- **Extra Practice 3, items 6–10:** 6 most accurate core = **R** · 7 affected by food/fluids = **O** · 8 rapid changes near the hypothalamus = **TY** · 9 "non-touch" = **TE** · 10 most unreliable = **A** (axillary).
- **First-quarter scope for abnormal temp: recheck** (same or another route), **notify** the primary nurse or provider, and document. With the nurse's OK you can do comfort measures: remove extra blankets, apply a cool cloth, offer oral fluids if there's no restriction (warm blankets for low temp). **No antipyretics**, because that's medication administration.

Source: Open RN *Nursing Skills* Ch. 1 Vital Signs; manufacturer instructions.

6. Respiration terms & Extra Practice 1 blanks

- **Orthopnea** = difficulty breathing **lying down**, relieved by sitting up (the blank on Slide 16).
- **Cheyne-Stokes** = cycles of breaths that get deeper then shallower, followed by **periods of apnea** (seen near death, with brain injury or heart failure).
- **Rales (crackles)** = fine popping/crackling sounds, usually on inspiration, from **fluid in small airways or alveoli**.
- **Documentation order: T, P, R, BP**, then SpO_2 and pain. Note the route and anything unusual.
- **Ivan example:** 13 beats × 6 = **P 78**; **R 16**; **T 100.6 °F oral**, which is a fever by her cheat sheet (>100.4). Write it as "T 100.6 °F (O), P 78, R 16."
- **SOB/dyspnea:** high-Fowler's (about 90°) or orthopneic (leaning forward over the overbed table) lets the lungs expand better. **O_2 is a medication and needs a provider order**, so a student reassesses and notifies.

Source: Open RN *Nursing Skills* Ch. 1 Vital Signs & Ch. 11 Respiratory Assessment.

7. Electronic VS machine (e.g. Welch Allyn Spot/Connex)

1. Hand hygiene; **clean the machine**; identify the patient and explain.
2. BP: choose the cuff by arm circumference and wrap it over the brachial artery (match the artery marker), with the arm at heart level. Press **Start** and have the patient keep still and not talk.
3. SpO_2: put the probe on a **warm finger with no nail polish** and wait for a steady waveform/pulse reading. The machine's HR is pulled from the probe.
4. Temp: put on a **probe cover**. **Blue probe = oral/axillary, red probe = rectal.** Wait for the beep.
5. **Check any abnormal reading manually**, and always check an irregular pulse manually (count apical for 1 full min). Count RR yourself, because the machine doesn't measure it.
6. Document, then clean the equipment and return it to the charger.

Source: Welch Allyn Connex Spot / Spot Vital Signs user manuals; Open RN *Nursing Skills* Ch. 1.

8. Manual 2-step BP (palpated systolic, then auscultated)

1. **Prep:** no caffeine, exercise or smoking for 30 min; empty bladder; seated with the **back supported, feet flat, legs uncrossed**; **rest 5 min**; bare arm supported at **heart level**; no talking.
2. **Cuff:** right size; bladder centered over the **brachial artery**; lower edge about **2–3 cm (1 in) above the antecubital fossa**.
3. **Step 1, palpated systolic (PSP):** feel the radial (or brachial) pulse and inflate until it **disappears**. That number is the estimated systolic. Deflate fully and **wait 30–60 sec**.
4. **Step 2, auscultate:** put the stethoscope over the brachial artery and inflate to **30 mmHg above the PSP**. Deflate at **2–3 mmHg per second**.
5. **First clear Korotkoff sound = systolic**; **sounds disappear = diastolic**. Keep listening 10–20 mmHg below the last sound, then deflate fully.
6. Record in **even numbers** (her gauge rule) with the arm and position. Wait **1–2 min** before repeating. Recheck and notify if abnormal.

Source: AHA Scientific Statement on BP Measurement (Muntner et al., *Hypertension* 2019); 2017 ACC/AHA BP guideline.

9. ABC prioritization

- **A**irway → **B**reathing → **C**irculation. A problem that threatens life now comes first. Other rules: **acute over chronic**, **actual over potential**, and physiological needs first (Maslow).
- **Her Slide 25 case (Jose):** by ABC, **RR 28** (breathing) is first; then circulation: **P 101** and **BP 165/94**. **T 101.8 °F** also needs follow-up (cellulitis, so an infection is spreading). Recheck, check SpO_2, and notify.

Source: standard NCLEX prioritization frameworks (NCSBN clinical judgment model; Open RN *Nursing Fundamentals* Ch. 4).

B. Vocab terms she lists but doesn't define

Her materials list or picture these terms but don't define them. These are standard nursing definitions, not hers. Check them against class notes or ask her.

Term	Meaning	Where she uses it
Virulence	How strong a germ is at causing disease.	Listed, Asepsis S15
Incubation period	Time from getting the germ until symptoms start.	Listed, Asepsis S17
Prodromal stage	Early, vague symptoms (tired, low fever) when the person is most contagious.	Listed, Asepsis S17–18
Full stage of illness	Specific symptoms of the illness are present.	Listed, Asepsis S17
Center of gravity / base of support	Center of a body's weight / the area under the feet you stand on.	Used, Activity S3, S7
Gait belt	Belt around the patient's waist that the nurse holds to steady them while walking or transferring.	Listed, Activity S11
Supine / prone / lateral / Sims'	Flat on back / flat on stomach / on the side / on the left side with right knee bent up, a half-prone position.	Pictured, Activity S12
Trendelenburg / reverse Trendelenburg	Bed tilted head-down / bed tilted head-up.	Pictured, Activity S12
Orthopneic position	Sitting up, leaning forward on the overbed table.	Pictured, Activity S12
Dangling	Sitting on the edge of the bed with legs over the side before standing.	Listed, Activity S13
Antipyretic	A fever-reducing drug (e.g., acetaminophen).	Listed, VS S9
Orthopnea	Trouble breathing when lying flat (her slide gives the description but leaves the term blank).	Blank, VS S16
Orthostatic hypotension	A drop in BP when standing up from lying or sitting.	Listed, VS S19
PQRST	Pain assessment: Provokes/Palliates, Quality, Region/Radiation, Severity, Timing.	Listed, VS S24
Closed-ended question	A question with a yes/no or one-word answer.	Listed, Comm S33
Helping relationship phases	Orientation → working → termination.	Listed, Comm S22